Egor Polyakov

Postdoctoral Researcher in Computational Musicology

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Profile

Computational musicologist, producer, and research software developer, connecting studio practice with analytical methods. My work combines production, mastering, composition, and the artistic-technical realisation of concerts and installations with teaching, project supervision, and computer-assisted music analysis. Practice-based knowledge shapes my research tools and my teaching. I am currently responsible for research and development for the DFG-funded CAMAT project at HfM Weimar.

Practice-to-Research Trajectory

Production	Ableton Live, Logic Pro, and Adobe Audition; mastering; Ambisonics with IRCAM Spat
Composition	Classical and electroacoustic composition, fixed media, live electronics, and multichannel work
Systems	Max/MSP, Max for Live, Pure Data, OpenMusic, analogue and digital hardware, and interactive workflows with sensors and controllers
Realisation	Teaching, project supervision, and the artistic-technical realisation of concerts, installations, and productions
Computation	Python, music encoding, reproducible audio and score analysis, and machine learning
Current research	Practice-informed computational musicology, open research software, and infrastructure

Integrated Competencies

Production and mastering	Recording, sound design, and mastering in Ableton Live, Logic Pro, and Adobe Audition; analogue and digital synthesis; studio signal flow and hybrid hardware/software setups; spatial production with Ambisonics and IRCAM Spat
Composition and artistic practice	Electroacoustic composition; fixed-media, live-electronic, multichannel, and installation formats; work across contemporary classical and electronic music
Interactive music systems	Max/MSP, Max for Live, Pure Data, and OpenMusic (Lisp); real-time audio, live electronics, sensors and controllers, algorithmic composition, and analysis–synthesis workflows
Concert, installation, and spatial-audio realisation	Artistic-technical realisation, rehearsal and system setup, live electronics, multichannel and spatial-audio system design and operation, restoration, and stem mastering
Computational musicology	Python, MEI/MusicXML, music21, and Verovio; corpus analysis, pattern discovery, and reproducible notebooks
Audio analysis and AI	NumPy, SciPy, pandas, scikit-learn, librosa, and PyTorch; timbre, rhythm, structure, embeddings, and generative modelling
Teaching and studio leadership	Research-led course design, project supervision, electroacoustic studio coordination, and technical mentoring
Research leadership and infrastructure	Grant development, team supervision, open science, Docker, cloud services, and Jupyter4NFDI

Languages

German professional fluency
English professional fluency
Russian native speaker

Professional Experience

2025–present **Postdoctoral Researcher**, *University of Music FRANZ LISZT Weimar*, Weimar

- Research and development for the DFG project *Development of a Comprehensive Cloud-Based Toolbox for Music Score Analysis*, alongside project head Prof. Dr. Martin Pfleiderer; responsible for scientific conception and research execution. Co-applicant; total funding: €443,458.
- Coordinate the research programme and supervise student assistants across technical and scholarly work packages.
- Develop Python workflows for MEI/MusicXML analysis, extend mei-friend for OMR corpora (musiconn / *Denkmäler deutscher Tonkunst*), and deliver interactive score workflows through MEI, Verovio, and Jupyter4NFDI.
- Integrate project methods into national research infrastructure and university teaching.

2013–2025 **Researcher and Artistic Associate**, *University of Music and Theatre Leipzig*, Leipzig

- Designed and taught courses in electroacoustics, popular and electronic music, computer-assisted analysis, and AI for musicology; supervised electroacoustic composition projects.
- Supported student productions and realisations artistically and technically, from recording and sound processing to live electronics, video, sensors, and controller integration.
- Coordinated the electroacoustic studio and maintained its analogue, digital, and live-electronics infrastructure.
- Realised and supported electronic works for concerts and installations, including rehearsal, system setup, multichannel and spatial-audio playback, restoration, and stem mastering.
- Developed tools spanning synthesis (FFTimbre), visualisation (AudioSpylt), and AI-assisted analysis across multiple repertoires.

2021–2022 **Research Associate**, *University of Music FRANZ LISZT Weimar*, Weimar

- Concurrent with the Researcher and Artistic Associate role at HMT Leipzig.
- Developed and evaluated CAMAT research and teaching pilots with faculty and students, including workshops on computational score analysis.

Career Context

Family context I am married and have two children. Between 2018 and 2022, family responsibilities and COVID-19 restrictions reduced the time available for conference travel and public-facing artistic production. I nevertheless remained professionally active at HMT Leipzig and, from 2021 to 2022, in the CAMAT research appointment at HfM Weimar.

Education

2025–present **Cumulative Habilitation (in progress)**, *University of Music FRANZ LISZT Weimar (HfM Weimar)*

2013–2018 **PhD in Musicology (Dr. phil.)**, *University of Music and Theatre "Felix Mendelssohn Bartholdy" Leipzig (HMT Leipzig)*

Dissertation on computer-based analysis and visualisation of music; supervisors: Prof. Dr. Gesine Schröder and Prof. Dr. Martin Supper.

2011–2014 **Master's studies in Composition and Electroacoustic Music**, *State University of Music and Performing Arts Stuttgart (HMDK Stuttgart)*

Studied with Prof. Marco Stroppa.

2010–2013 **Postgraduate Artist Diploma (Meisterklasse) in Electroacoustic Music**, *University of Music and Theatre "Felix Mendelssohn Bartholdy" Leipzig (HMT Leipzig)*

Prof. Ipke Starke.

2004–2010 **Diploma in Composition**, *University of Music and Theatre "Felix Mendelssohn Bartholdy" Leipzig (HMT Leipzig)*
Profs. Peter Herrmann and Ipke Starke.

Teaching

More than twelve years of teaching across electroacoustic composition and production, popular and electronic music, computer-assisted analysis, digital humanities, and AI at HMT Leipzig (2013–2025) and HfM Weimar (since 2025), combining artistic practice with research-led technical instruction.

Selected courses

WS 2026/27 "Von der digitalen Edition zur computergestützten Musikanalyse: Einführung in Musik-Encoding und Tools für die Notentextanalyse", HfM Weimar – Introductory seminar on music encoding and digital tools for computational score analysis.

SS 2025 "Hearing (and analyzing) in time", HMT Leipzig – Groove and metric analysis in electronic music.

WS 2024/25 "Algorithmic music models: from analysis to style imitation with Python/AI", HMT Leipzig – Probabilistic and machine-learning techniques.

WS 2023/24 "AI and statistical analysis in musicological research", HMT Leipzig – Score and audio pipelines with machine learning.

SS 2023 "From track to DJ set, from sample to live performance", HMT Leipzig – Macroform concepts in popular electronic music.

2013–2025 "Fundamentals of Electroacoustic Music I and II", HMT Leipzig – Annual foundational seminar covering acoustics, digital production, and live-electronics practice.

Artistic and Technical Support (Selection)

2018 **"die maschine steht still" (Johanna Wokalek, Fabian Russ)**
- **Role:** live electronics realisation, stem mastering.
- Premiere — Futurium Berlin

2017 **"Black Is the Colour" (Fabian Russ)**
- **Role:** restoration and mastering for CD and online release.
- Label: Neue Meister (Edel)

2016 **"Für Tuba mit Hegel" (Georg Katzer)**
- **Role:** rehearsal, live electronics, restoration.
- University of Music and Theatre Leipzig

2015 **"Butterfly under glass" (Fabian Russ, Laurie Young, Frieder Weiss)**
- **Role:** live electronics and stem mastering.
- Premiere — Scala Esslingen

2015 **"Harmonia Mundi" (Fabian Russ)**
- **Role:** live electronics realisation, stem mastering.
- Premiere — Montforthaus Feldkirch

2013 **"Inside Partita" (Folkert Uhde, Midori Seiler, Fabian Russ)**
- **Role:** live electronics realisation, stem mastering.
- Premiere — St. Elisabethkirche, Berlin

2009 **"Utopia" (Thomas Kessler)**
- **Role:** rehearsal and technical assistance.
- Premiere — Viehauktionshalle Weimar

Mastering

Mastering for the Ornithopter Records catalogue as part of the label's founding team, and for releases by bands and underground artists from Leipzig and Berlin, alongside pre-mastering for Fabian Russ's contribution to *Neue Meister: Live in Berlin* (Neue Meister / Berlin Classics / Edel:Kultur). Full mastering discography and credits.

Artistic Works (Selection)

2013 **"Core v2"**
for oboe, English horn, and 8-channel live electronics

2012 **"Manuals"**
for Paetzold recorder quartet and 4-channel live electronics

2011 **"Core v1"**
for oboe, English horn, and 4-channel live electronics

2011 **"Stereo-Type II"**
for 2-channel tape

2010 **"Stereo-Type"**
for 2-channel tape

2009 **"Double Helix"**
for oboe, cello, and 4-channel live electronics

2008 **"Schrittmacher"**
4-channel sound installation

Releases (Electronic Music)

2009 **Two Unknown Guys – "Swirl Planet EP"**
Ornithopter Records (OR 005)

2008 **Two Unknown Guys – "The Rubber Duck Massacre EP"**
Sinergy Networks (SN056)

Externally Funded Projects

2025–2028 ***Development of a Comprehensive Cloud-Based Toolbox for Music Score Analysis***
- DFG grant (€443,458) — co-applicant; research and development; project head and organisational lead: Prof. Dr. Martin Pfleiderer.

2021–2022 ***Computergestützte Musikanalyse***
- Fellowship funding (TMWWDG / Stifterverband) secured by Prof. Dr. Martin Pfleiderer, who led the project. Contributed as scientific assistant to CAMAT research, development, and curricular evaluation.

Publications

Published

2026

Hentschel, J. et al. [20 authors, incl. Polyakov, E.] (2026). *Time to Align! Modelling Musical Timelines for Music Information Retrieval and Digital Musicology*. *Transactions of the International Society for Music Information Retrieval* 9(1), 384–404. DOI: 10.5334/tismir.296. DOI

2026

Polyakov, E. (2026). *FFTimbre: Exploration of Timbre by Analysis and Synthesis Using Python, Ableton, and Large Language Models*. In: *Innovation in Music: Current Research Perspectives* (pp. 121–134). Focal Press / Routledge. DOI: 10.4324/9781003475675-11. DOI

2026

Polyakov, E. (2026). *eu-pd-composers: EU public-domain estimates for 20th-century classical composers and their works on IMSLP* [Dataset]. Zenodo. DOI: 10.5281/zenodo.23730119. DOI

2025

Polyakov, E. (2025). *Encoding the New Frontier: Adapting MEI and Verovio for Post-Tonal and Spectral Notations*. In: *Music Encoding Conference 2025 Book of Abstracts*. DOI: 10.17613/exk9y-s3x74. DOI

2024

Pfleiderer, M.; Polyakov, E.; Nadar, C. (2024). *Analyse! Development and integration of software-based tools for musicology and music theory*. In: *Innovation in Music: Technology and Creativity* (pp. 292–309). Routledge. DOI: 10.4324/9781003118817-21. DOI

2021

Polyakov, E. (2021). Articles on George Crumb, loudspeaker orchestras, orchestras and new media, and recording technology. In: *Lexikon des Orchesters*. Laaber-Verlag. Link

2020

Polyakov, E. (2020). *Computerbasierte Analyse und visuelle Repräsentationsformen der Musik*. Dissertation, University of Music and Theatre Leipzig. Link

2019

Polyakov, E. (2019). *«Крайне затруднительны и не точны» — описание инструментальных тембров в «Основах оркестровки» (1913) Н. А. Римского-Корсакова* ["Extremely difficult and inaccurate": instrumental timbres in Rimsky-Korsakov's *Principles of Orchestration* (1913)]. In: *Год за годом. Римский-Корсаков — 175. Материалы международной научной конференции 18–21 марта 2019 г.* (pp. 37–51). St Petersburg: СПб ГБУК СПбГМТиМИ. Link

In Press / Accepted

in press

Polyakov, E. (in press). "Understanding and Emulating Time: Analyzing and Simulating Musical Microrhythm Timing with the beat_it Toolbox". In: *Innovation in Music: Sound and Place*. Focal Press / Routledge. Link

in press

Polyakov, E. (in press). "Evaluation of Modern Computer-Aided Sheet Music Analysis Methods in a Practical Context". In: *GMTH Proceedings*.

Under Review

submitted

Polyakov, E. "Large Language Models as Access Layers for a Cloud-Based Music Workflow: An Exploratory Case Study". Submitted to *Proceedings of Innovation in Music 2026* (InMusic26). · Materials

Talks and Conferences

2026 (presented) **Poster: "CAMAT_v2: Cloud-Based Tools for Collaborative Music Score Analysis"**
ICCCM26, Würzburg, 21–23 September 2026

2026 (presented) **Talk: "Beyond the Black Box: Democratizing Musical Analysis and Creative Workflows via LLM-Empowered Cloud-Based Jupyter Setups"**
InMusic26, Aalborg, Denmark, 12–14 June 2026 (with P. Steuck) · Materials

2025 **Talk: "How Machines Listen"**
25th Annual Congress of the Society for Music Theory (GMTH), Lübeck, 17–19 October 2025

2025 **Symposium organisation: "Challenges of Computer-Assisted Score Analysis"**
Society for Music Research (GfM) annual conference, Weimar, 6–9 October 2025

2025 **Talk: "Binary score representations — an interactive method for motif, chord, function, and texture search in symbolic score representations"**
GfM annual conference, Weimar, 6–9 October 2025

2025 **Talk: "Understanding and Emulating Time: Analyzing and Simulating Musical Microrhythm Timing with the beat_it Toolbox"**
Innovation in Music 2025, Bath (UK), 20–22 June 2025

2025 **Poster: "Encoding the New Frontier: Adapting MEI and Verovio for Post-Tonal and Spectral Notations"**
Music Encoding Conference 2025, London, 3–6 June 2025

2024 **Workshop: "AI for Music Theory"**
24th Annual Congress of the Society for Music Theory (GMTH), Cottbus, 4–6 October 2024

2024 **Talk: "How Constant Is Your Beat? Computer-Assisted Analysis of Beat and Tempo Fluctuations from Acousmatic Music to Minimal Techno with the beat_it Toolbox"**
Rhythm under the Microscope: Microrhythm and Groove in Popular Music, mdw Vienna, 25–27 September 2024

2024 **Talk: "Exploring Electroacoustic Music Analysis with Multimodal Large Language Models"**
GfM annual conference, Cologne, 11–14 September 2024 · Materials

2023 **Talk: "Reconstruct and Decompose: Extracting and Sonifying Rhythmic, Melodic and Spectral Patterns for Analytical and Creative Use"**
GMTH annual conference, Freiburg, 22–24 September 2023

2022 **Workshop: "Evaluation of computer-assisted music score analysis methods within the Fellowship project Computer-assisted music analysis at music university Weimar"**
GMTH annual conference, Salzburg, 30 September–2 October 2022

2022 **Talk: "Analyze! Development and integration of software-based tools for musicological and music theoretical needs"**
Innovation in Music Conference 2022, Stockholm, 17–19 June 2022 (with M. Pfleiderer and C. Nadar)

2019 **Talk: "Immer ungenau und mangelhaft — description of instrumental tone colours in Rimsky-Korsakov's 'Principles of Orchestration' in the context of current developments in computer-based timbre analysis"**
Rimsky-Korsakov at 175, St. Petersburg, 18–21 March 2019

2017 **Talk: "Self-similarity matrix in the context of computer-based timbre analysis and its use in teaching"**
Symposium: Popular Music and its Theories, GMTH annual congress, Graz, 17–19 November 2017

Professional Memberships

since 2016 Gesellschaft für Musiktheorie (GMTH)

since 2018 Deutsche Gesellschaft für Elektroakustische Musik (DEGEM)

since 2024 Gesellschaft für Musikforschung (GfM)

Note Born in Bakhmut, Ukraine. Based in the Leipzig region, Germany. Publications may appear under the former transliteration "Poliakov".